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Search Algorithm

ERP Assignment - 1

1) Define following terms 1) search space 2) path cost 3) Transition Model 4) optimal solution

A:- 1) search space:
search space is the set of all possible states or solutions that can be explored to solve a problem. It contains all possible paths from the initial state to the goal state.

2) path cost:
path cost is the total cost of a path from the initial state to a goal state.
It is calculated by adding the cost of each step taken along the path.

3) Transition model:
Transition model describes how one state changes to another state after applying an action.
It defines the result of performing a particular action in a given state.

4) optimal solution:
An optimal solution is the best possible solution among all solutions.
In search problems, it is the solution with the lowest path cost.

2) Diffe

2, Distinguish between an uniformed search algorithm and an informed search algorithm based on their reliance on domain knowledge.

uniformed search	Informed search
• Does NOT use domain knowledge	• uses domain knowledge
• Explores blindly without knowing goal direction	• uses heuristic to guide search toward goal
• usually slower and explores more nodes	• Faster and explores fewer nodes
• Ex: BFS, DFS	• A* search, Greedy Best-First search
• Not focused in goal	• Focused in goal